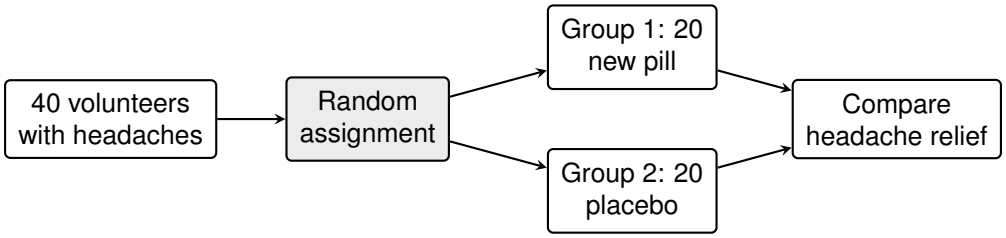
# Experiments KEY

Name _____

Does a new pill stop headaches? We cannot just compare people who chose to take it with people who did not. They may be different in other ways. An **experiment** gives the treatment on purpose and compares fairly.

| Word | In the headache experiment |
|---|---|
| Subjects (experimental units) | The 40 volunteers with headaches |
| Explanatory variable | What we change: the type of pill |
| Treatments | The things we compare: new pill, placebo |
| Response variable | What we measure: headache relief |
| Control group | The group that does not get the new treatment (the placebo group) |
| Placebo | A fake treatment that looks real, like a sugar pill |
| Double-blind | Neither the subjects nor the people measuring know who got which pill |




**Principles of a good experiment**

**Control** Compare to a control group. Keep everything else the same for both groups.

**Random assignment** Use chance to put subjects into groups. Then the groups are alike except for the treatment.

**Replication** Use enough subjects that chance differences between people even out.

**Why a placebo?** People often feel better just because they *think* they got medicine (the **placebo effect**). With a placebo group, both groups think they are being treated.

**Matched pairs design.** Each subject gets **both** treatments, in a random order. Or pair up very similar subjects, like twins.

**Example:** Which running shoe is faster? Each runner runs once in each shoe. A coin flip decides which shoe goes first. Comparing each runner to themselves removes the differences between runners.

Work with a partner. Compare answers when you finish each problem.

1. 60 tomato plants are randomly split into two groups of 30. One group gets a new fertilizer. The other gets only water. After 6 weeks, the weight of the tomatoes is measured.

   (a) What are the subjects? The 60 tomato plants

   (b) What is the explanatory variable? Fertilizer or water (the treatment)

   (c) What is the response variable? The weight of the tomatoes

   (d) What is the control group? The 30 plants that get only water

   (e) Why split the plants **randomly**? So the groups are alike. Then a difference comes from the fertilizer, not from which plants were stronger.

2. In the headache experiment, the study is double-blind.

   (a) Why should the subjects not know which pill they got?
   If they know they got the real pill, they might feel better just from believing it (the placebo effect).

   (b) Why should the nurses who measure headache relief not know?
   They might, without meaning to, rate the new-pill group better.

3. Does listening to music help students memorize a list of words? You have 50 student volunteers. Describe a completely randomized experiment. Name the treatments, how you assign them, and what you measure.

   Treatments: music, no music. Number the students 1 to 50 and use a random number generator to put 25 in each group. Both groups study the same list for the same time in the same room. Measure how many words each student remembers. Compare the two groups.

4. A company tests two hand lotions. Each of 20 volunteers puts Lotion A on one hand and Lotion B on the other. A coin flip decides which hand gets A. After a week, a doctor rates the dryness of each hand.

   (a) What kind of design is this? Matched pairs

   (b) Why flip a coin to pick the hand? Most people use one hand more. The coin keeps that from favoring one lotion.

   (c) Why is this design better than giving 10 people Lotion A and 10 people Lotion B?
   Each person is compared to themselves, so differences in skin from person to person do not matter.

5. A teacher tries a new reading program with her 1st-hour class and the old program with her 6th-hour class. The 1st-hour class scores higher. Can she conclude the new program works better? Explain.

   No. She did not use random assignment. The classes could differ in other ways (time of day, which students take 1st hour). Those differences could cause the higher scores.